

A Rare Comet Made History as the Third Known Interstellar Object to Fly Through Our Solar System. Studies Are Now Revealing the Mysterious Conditions in Which It Formed

Genre: Science | 1419 words

Source: <https://www.smithsonianmag.com/science-nature/a-rare-comet-made-history-as-the-third-known-interstellar-object-to-fly-through-our-solar-system-studies-are-now-revealing-the-mysterious-the-mysterious-conditions-in-which-it-formed-180989037/>

A Rare Comet Made History as the Third Known Interstellar Object to Fly Through Our Solar System. Studies Are Now Revealing the Mysterious Conditions in Which It Formed

In the latest research, telescope observations of 3I/ATLAS examine its chemical structure and suggest that it took shape in a peripheral region of a relatively calm planetary system

After a visit from beyond the solar system, 3I/ATLAS, the third interstellar object ever detected, is on its way out. Distinct from anything that circles the sun, the icy body has now passed the orbit of Jupiter. Soon it will slip back into the void, never to be seen by humanity again.

The object—a cometlike body enshrouded in gas and dust—was forged in the debris disk surrounding a distant star, providing an extraordinary opportunity for experts to study the building blocks of another planetary system. Astronomers around the world have trained telescopes on this celestial visitor ever since its discovery on July 1, 2025. Both before and after its closest approach to the sun, when the heat of our star vaporized material from its surface, they have analyzed the comet's composition.

Key context: A trio of interstellar objects

3I/ATLAS is the third interstellar object discovered in our solar system. The first, 1I/'Oumuamua, is a cigar-shaped object discovered in 2017. 2I/Borisov was spotted in 2019, and it was the first interstellar object to be confirmed as a comet.

After a year of observation and study, scientists now say that 3I likely originated in the cold outer reaches of its home star's protoplanetary disk, a ring of debris that later coalesced into objects like planets, moons and asteroids.

The latest of this work, posted Sunday on arxiv.org as a preprint awaiting peer review, used the James Webb Space Telescope to look at the dust of 3I. Based on the structure of silicate compounds in the comet's coma, the team suggests that 3I formed far away from the inner part of its protoplanetary disk. Along with other chemical clues from hydrogen and carbon isotopes, the new observations add to a growing body of research suggesting that 3I is from the outskirts

of an ancient system.

“It’s making a really nice picture, all of these pieces of evidence, that we have an unusual object that formed really far from its host star,” says Karen Meech, an astronomer at the University of Hawaii. “I don’t know that we’ve seen such a creature in our solar system.”

Crystalline clues to a far-out origin

Most of the mass in rocky bodies across the galaxy—terrestrial planets, asteroids and comets—consists of silicates, or minerals of silicon and oxygen. These solid compounds can be either crystalline, meaning they have an ordered atomic structure, or amorphous, meaning the atoms don’t line up in any particular way.

Planets, moons and smaller rocky bodies arise from the protoplanetary disk that surrounds their central star. By modeling this formation process, scientists know that crystalline silicates take shape most readily in the inner part of the disk, closer to the star. In our own solar system, these crystalline silicates were then swept to the outer regions by great streams of gas and dust flowing outward from the sun.

“Early on, before the gas and the dust dissipated, the solar system had these turbulent, large-scale flows,” says Matthew Belyakov, an astronomer at Caltech who worked on the new 3I dust analysis. “That turbulence in the solar system was high enough where material from the inner part of the system managed to mix outwards.”

As a result, comets that formed in the outer regions of our solar system still contain crystalline silicates. “3I does not look like that,” Belyakov says.

Using the Mid-Infrared Instrument on Webb, Belyakov and his colleagues analyzed a signature from the light coming from the dust surrounding 3I. By splitting the light into a spectrum, the researchers could assess what portion of 3I’s silicates are crystalline in structure.

The answer was almost none. “It didn’t get any of that material that condensed near its host star,” Belyakov says. “It really just formed from the dust that came from the outside.”

“To me this was surprising,” says Meech, who was not involved in the research. “There clearly wasn’t much mixing, because it didn’t really have much in the way of crystalline silicates.”

It remains possible that some crystalline silicates were hidden from the telescope by the glow of large dust grains, which might have masked their signal. Still, the amorphousness of 3I hints that its home environment was quiet—much less turbulent than the solar system we know—indicating that other planetary systems could be strikingly different from our own.

A deep freeze

Additional hints of 3I's birthplace come from its chemical composition. These include that 3I has elevated levels of carbon monoxide, carbon dioxide and methane compared with solar system comets. These compounds readily burn off in warm conditions, pointing to a cold formation environment.

Both Webb and the Atacama Large Millimeter Array (ALMA), in Chile, were also able to tease apart the water molecules in the gas of 3I's coma. What they found was an abundance of so-called heavy water.

Although a normal water molecule contains an oxygen atom and two hydrogen atoms, water can also comprise an oxygen atom, a regular hydrogen atom and a heavier variety of hydrogen called deuterium. Measurements from ALMA, published in April, indicate that 3I's ratio of heavy water to normal water is about 30 times higher than what's seen in our solar system's comets, providing additional evidence of frigid formation conditions—below minus 405 degrees Fahrenheit.

"When the temperatures are low, only the reaction that increases deuterium can happen," says Luis Manzano, an astronomer at the University of Michigan who worked on the ALMA observations. "Our expectation is that 3I/ATLAS was formed in the outer reaches of its protoplanetary disk, because the farther you are from your protostar, the lower the temperatures you can get."

Another ratio, the amount of carbon-12 to the heavier carbon-13, also reveals 3I's unique traits. According to measurements from Webb published in June, the comet has a low level of carbon-13 compared with objects in the solar system and nearby interstellar clouds and protoplanetary disks. This hints that 3I may have formed in the ancient past, as many as 12 billion years ago, possibly around a star on the fringes of the Milky Way, where there are not as many heavy elements.

"Outflows from large stars will generate more carbon-13 and slowly populate the galaxy with [more] carbon-13 relative to carbon-12," says Belyakov, who was not involved with the carbon findings. "3I does not have very much carbon-13, so that is used to indicate that it formed early on in our galaxy's history. It's not definitive evidence of that, but it's as solid evidence as any."

Taken together with trajectory estimates that 3I flew through interstellar space for billions of years before reaching us, this ratio could mean that the roughly 1.5-mile-wide chunk of rock and ice is the oldest planetary body ever seen.

Chasing down cosmic interlopers

Although telescopes on Earth and in space have revealed extraordinary details about 3I/ATLAS, there are still unknown variables that would provide even more information about the place it came from.

“Measurements of oxygen isotopes, nitrogen isotopes, noble gases—these are all different fingerprints of processes that go on in the disk,” Meech says. “It’s the isotopic work that I think would give us really good clues as to where it formed and what sort of physics is going on, but some of these measurements would be very difficult to do from Earth.”

To get up close and personal with a future interstellar object, some scientists have advocated for an interceptor spacecraft to perform a flyby. This would be challenging, but not impossible. In 2028 or 2029, the European Space Agency plans to launch the Comet Interceptor mission, which will park in space and wait for an undiscovered long-period comet to target. If we get lucky, perhaps another interstellar object will visit while the Comet Interceptor is waiting. “If you could fly a mass spectrometer through its dust and get all these detailed measurements ... those experiments would be spectacular,” Meech says.

Scientists don’t know how often interlopers visit from beyond the solar system, but when the next ones arrive, telescopes such as the new Vera C. Rubin Observatory, which revealed its first photos last year, have a good chance of spotting them.

“We might enter a golden age of interstellar object astronomy in the next few years,” Belyakov says.